

Medical retina and macular decisions

Parts 1 and 2 | Describe before prescribing

Separate structure, activity and the source of lost function.

Finding	Implication
What is the space?	Fluid, degeneration, splitting or traction
Where is the lesion?	Retinal, subretinal or sub-RPE compartment
What is changing?	Acuity, symptoms, haemorrhage and serial imaging
What can improve?	Active mechanism versus irreversible damage

With distortion but preserved letter acuity, OCT may show tangential membrane traction rather than leaking cysts. Compare serial surface deformation, outer-retinal integrity and the patient's reading symptoms. A thin retina with degenerative cavities requires a different interpretation from a swollen leaking retina; an OCT space alone cannot establish an injection indication.

A trial result has a population and a regimen

Identify the comparison before applying its headline.

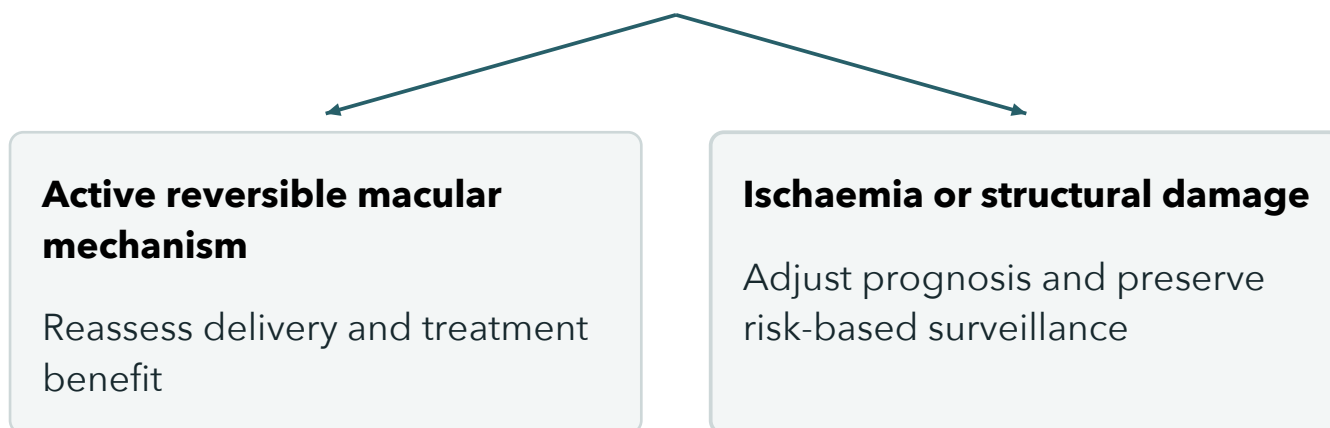
Finding	Implication
PCV trials	Drug and prompt-versus-rescue PDT differ
DME trials	Baseline vision and follow-up time matter
CSC trials	Mechanistic plausibility did not validate eplerenone
Translation	Name the endpoint and relevant exclusions

EVEREST II tested ranibizumab with prompt PDT; PLANET tested aflibercept with selective rescue PDT. Their different designs explain why their conclusions cannot be reduced to a universal PCV combination rule. Protocol T additionally shows that a comparative DME result changes with baseline acuity and assessment year; VICI did not validate eplerenone for chronic CSC.

Treatment response and residual risk

A central improvement can coexist with a different threat.

What remains after the scan improves?



After treatment of ischaemic CRVO, a dry macula and improved acuity do not establish recovered perfusion. Examine the iris and angle for neovascularisation and plan surveillance around injection cessation. Persistent low acuity despite reduced thickness may reflect ischaemia or structural injury, whereas residual fluid with ongoing functional benefit may still justify treatment.

Inherited disease: function anatomy and genotype

Three kinds of evidence answer three different questions.

Finding	Implication
Electrophysiology	Which function is impaired?
Imaging	Which tissue remains structurally viable?
Molecular result	Which causative variants are established?
Treatment discussion	Does the specific indication fit this eye?

A confirmed molecular diagnosis does not by itself establish gene-therapy eligibility. Voretigene requires biallelic RPE65-associated disease and viable retinal cells; a different inherited maculopathy does not meet that indication. Structural viability determines whether the target remains available, while functional testing describes impairment. Discuss realistic improvement and continuing rehabilitation rather than promising normal sight.

The endpoint should matter to the patient

Dry anatomy, acuity and useful daily vision are related but distinct.

Finding	Implication
Distortion	May be important despite good letter acuity
Acuity plateau	Review optics, perfusion and structural limits
Treatment burden	Attendance and adverse effects influence benefit
Shared decision	Define what improvement would be worthwhile

A symptomatic epiretinal membrane can impair reading through distortion despite good central acuity. Surgery addresses traction; it cannot restore retina already lost to atrophy or fibrosis. Conversely, active exudation offers a treatable vascular target. Define success using the relevant symptom and structure, since anatomical drying and improved daily visual function may diverge.